

Crypto Trading Model

Final Presentation - Team 28

Overview

Predict the long-term trend of cryptocurrency prices based on historical market data

Time horizons	15 mins, 1 hour, 4 hours, 1 day
Crypto coin pairs	USDTpairs: BTC/USDT, ETH/USDT, LTC/USDT, SOL/USDT BTCpairs: ETH/BTC, LTC/BTC, SOL/BTC
Target variables	Log_return_4h for all major coins
Input features	Raw market data (OHLCV)
	Technical indicators: VWAP, EMA, MACD, RSI, ATR, Bollinger Bands, OBV, log return

Backtest

Environment Specifications

- Assets: BTC/USDT, ETH/USDT, LTC/USDT, SOL/USDT
- Starting Capital: \$10,000
- Position Sizing: Maximum 40% per asset (based on starting balance)
- Fee Structure: Alpaca taker fees (0.25% per side, 0.5% round-trip)
- Data Frequency: 15-minute bars
- Test Period: Out-of-sample test set (most recent 15% of data)

Backtesting results

Sharpe Ratio Formula

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

R_p = return of portfolio

R_f = risk-free rate

σ_p = standard deviation of the portfolio's excess return

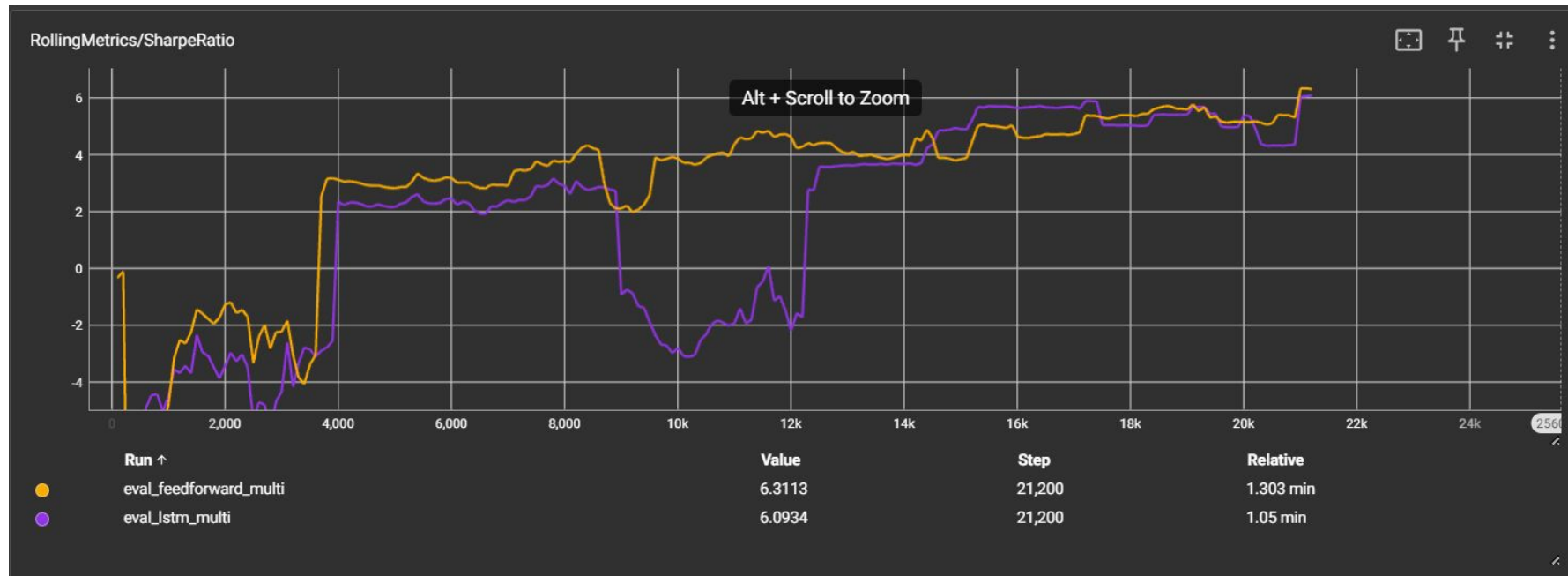
INSIDER

Information Ratio Formula


$$\text{Information Ratio Formula} = \frac{(R_p - R_b)}{\text{Tracking Error}}$$



Backtesting results



Validation results

8.5.1 Feedforward MLP Performance

Metric	Value
Starting Balance	\$10,000.00
Final Equity	\$321,463.48
Total Return	+3,114.63%
Sharpe Ratio	3.268
Max Drawdown	-45.24%
Win Rate	56.79%
Total Steps	21,247
Total Trades	166
Total Fees	\$1,649.50 (16.49%)
Avg Fee/Trade	\$9.94

Table 6: Feedforward MLP backtesting results

8.5.2 LSTM Performance

Metric	Value
Starting Balance	\$10,000.00
Final Equity	\$255,001.81
Total Return	+2,450.02%
Sharpe Ratio	2.977
Max Drawdown	-42.18%
Win Rate	54.21%
Total Steps	21,231 (seq buffer: 16)
Total Trades	142
Total Fees	\$1,421.33 (14.21%)
Avg Fee/Trade	\$10.01

Table 7: LSTM backtesting results

```
> python alpaca.py
```

Account Summary

```
cash: 38695.46
portfolio_value: 103362.9
equity: 103362.9
buying_power: 77390.92
```

PERFORMANCE METRICS

```
Sharpe Ratio:      3.255
Total Return:      4.38%
Max Drawdown:      -2.07%
Volatility (Annual): 14.92%
Win Rate:          0.00%
Average Win:        $0.00
Average Loss:       $-0.00
Profit Factor:      0.00
```

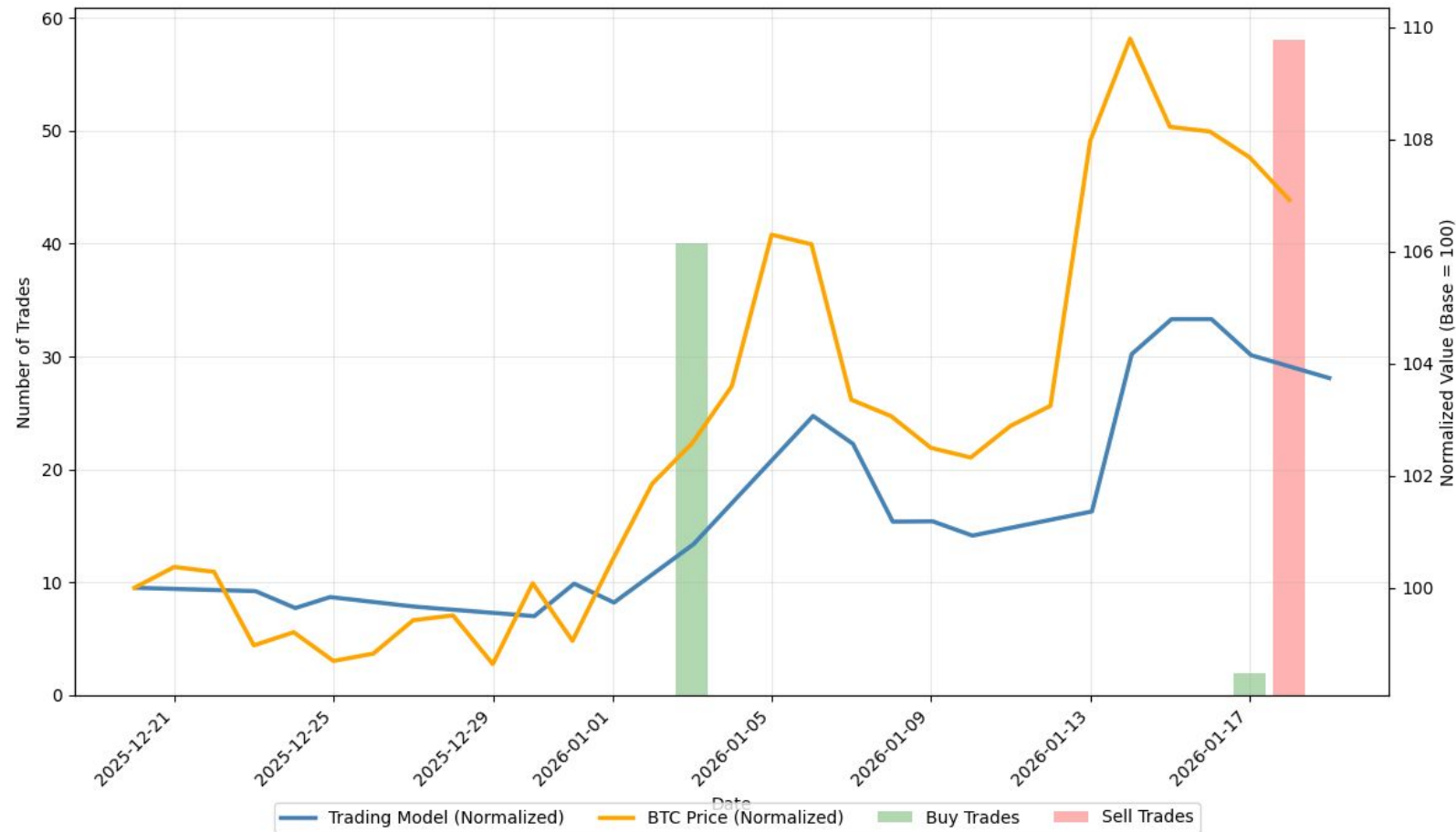
BTC BUY AND HOLD ANALYSIS

```
Sharpe Ratio:      3.514
Total Return:      10.34%
Max Drawdown:      -3.74%
Volatility (Annual): 22.34%
```

Paper Trading

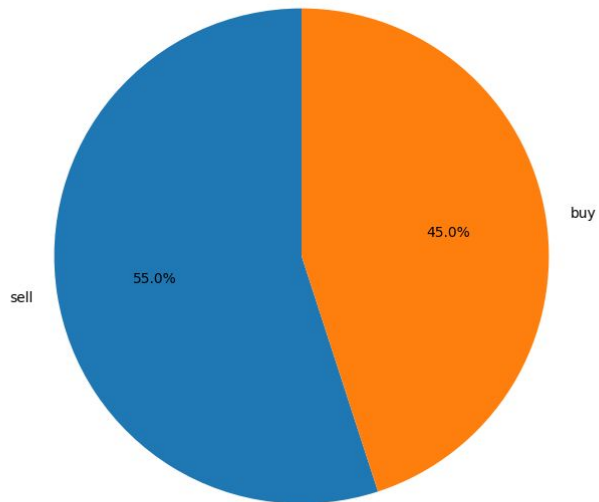
Paper Trading

Combined Analysis: Model Performance, BTC Price, and Trade Activity

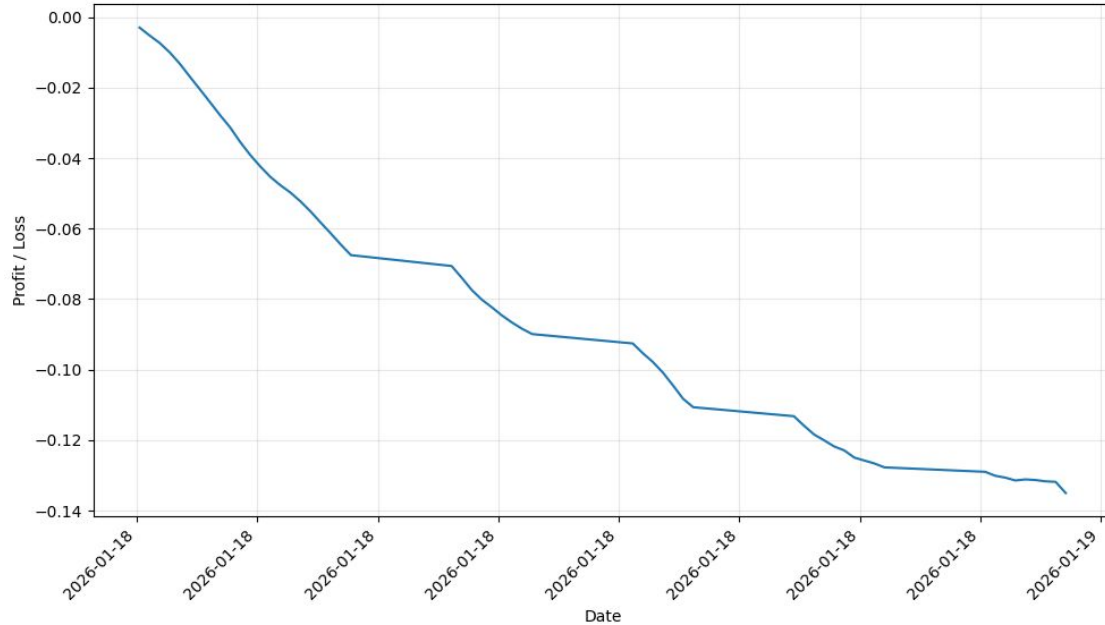


Paper Trading

Buy vs Sell Distribution



Cumulative Realized P/L (FIFO)



Paper Trading: Potential explanations of results

- low risk factor
- result limited by only trading 1 coin
- inconsistent timeframes for averages -> react to movements faster early

```
pred_history = deque(maxlen=100) # keep the last 100 predictions
if len(pred_history) >= 20: # Wait for at least 20 samples to be safe
    # raw_values = [p[1] for p in pred_history]
    # rolling_mean = sum(raw_values) / len(raw_values)
    raw_values = np.array([p[1] for p in pred_history])
    rolling_mean = raw_values.mean()
    rolling_std = raw_values.std()
```

Iterative improvements and Optimization

1. Feature Engineering & Normalization

- **Challenge:** Initial data scaling methods were inconsistent with the statistical properties of the features.
- **Solution:** Transitioned to **Layer Normalization** and converted raw price data to **Log>Returns**.
- **Result:** Achieved stationarity in the input distribution, leading to faster convergence and better generalization across different market regimes.

Iterative improvements and Optimization

2. Model Stability & Architecture

- **Challenge:** Observed high variance and "spiking" loss curves during the training phase, indicating gradient instability.
- **Solution:** Refined the architecture for both **Feed-Forward** and **LSTM** models, implementing gradient clipping and optimized weight initialization.
- **Result:** Stabilized the training loop, resulting in a smooth, predictable loss decay and improved model reliability.

Iterative improvements and Optimization

3. Execution Logic & Paper Trading

- **Challenge:** Fixed signal thresholds led to "over-trading" and poor performance in low-confidence environments.
- **Solution:** Developed a **Dynamic Thresholding** mechanism paired with an **execution buffer** to account for market noise.
- **Result:** Significant reduction in false-positive trades and improved net profitability by optimizing entry/exit precision.